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Department of Mechanical and Mechatronics Engineering

**Design of a Remote Arming
System for Sounding Rocket
Recovery Systems**

A Report Prepared For:
The University of Waterloo

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Dear Professor Kennings,

This report, entitled Design of a Remote Arming System for Sounding Rocket Recovery Systems, was written to fulfil my 3A work term report requirements. This is my 2nd work report submission.

Remote Arming is a system I have been developing since the fall of 2019 and is currently being prepared to fly on Waterloo Rocketry's newest rocket, Kraken of the Sky, at the inaugural Launch Canada competition in October.

This report was written entirely by me and has not received any previous academic credit at this or any other institution. The Remote Arming system has been entirely scoped, designed, assembled, and tested by me. I would like to acknowledge the help of members of the Waterloo Rocketry team in helping to identify requirements, reviewing design decisions and concept of operations, and providing code reviews for the Remote Arming code base.

Best Regards

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Summary

Each year the Waterloo Rocketry design team designs, manufactures, and flies a hybrid sounding rocket at the Spaceport America Cup and as of 2021 the team is also competing in the new Launch Canada competition. Currently the team is finishing its latest rocket, Kraken of the Sky, which will fly to an altitude of 30,000 ft, before deploying a sequence of parachutes which will allow it to touchdown softly and be recovered.

To deploy the parachute pyrotechnic charges are used. Since these pyrotechnics poses significant stored energy, the altimeters which control their timing are armed just before the last two personnel are evacuated from the launch site. Due to the height of the rocket, 18ft, to arm the magnetic switches, an operator must climb a step ladder, and wave a pole with a magnet attached to the end over the top of the rocket. Since this is somewhat precarious and has the added risk of placing the operator directly under the armed pyrotechnics, a better solution was desired.

To improve the safety of arming the recovery pyrotechnics the Remote Arming system was developed. The goal of this system was to use the rocket's pre-existing CAN bus and communication systems to control power to the altimeters. In addition, the system would collect telemetry data about the state of the recovery system to allow monitoring its health up to and including launch.

The key challenge when designing the Remote Arming system was ensuring the system was redundant and fail safe. This is particularly challenging because there is not a single safe state for Remote Arming. While on the ground the system must always fail disarmed to ensure there is no risk of personnel being injured by the pyrotechnics being inadvertently fired. However once in the air, the system must always fail armed, as otherwise it will prevent the parachute from deploying and will cause the rocket to impact the ground at incredible speed. Since always failing both armed, and disarmed is not possible, a compromise was made. The system is designed to always fail armed via a number of both hardware and software measures, however the same magnetic switches used to arm the recovery system in previous years are used to allow the system to be manually disarmed, and a pair of buzzers ensure that personnel cannot inadvertently approach an armed system.

The system was designed in a similar fashion to the rest of Waterloo Rocketry's RocketCAN boards, including the same microcontroller, and basic peripherals. It supports two redundant batteries, each with their own magnetic control switch. It has two separate altimeter control circuits, which use a flight proven circuit to provide power to each altimeter. In order to monitor the performance of the recovery system, Remote Arming has eleven different telemetry channels, which monitor parameters such as battery voltage and current draw, as well as measuring the voltage of the pyrotechnic igniters, allowing faults to be detected before launch, and allowing parachute deployment to be detected in flight. It also can communicate with the Stratologger altimeter to determine the rocket's current altitude. At present, the Remote Arming system has been built, and undergone significant development and benchtop testing, however there are still a few tests which must be completed before the system is ready for flight.

1.0 Introduction

The Waterloo Rocketry Team is a student design team consisting of approximately 50 members. Its primary goal is to design, build, and fly high powered sounding rockets [1]. Each year the team competes in New Mexico at the Spaceport America Cup, most recently placing 2nd in the 30,000ft hybrid/liquid division in 2019, with our rocket, shown in Figure 1, Shark of the Sky (SOTS). In addition to points awarded for presentation and technical report quality, the primary objective of the competition is to get as close as possible to the target altitude, and then successfully recover the rocket. A large fraction of the total points are awarded for successfully flying the rocket, and then recovering it in good condition such that it could be flown again with no repairs. Currently the team is working on completing the design and manufacturing of our next rocket, Kraken of the Sky (KOTS), a hybrid sounding rocket targeting 30,00ft which builds on many lessons learned from the launch of SOTS.

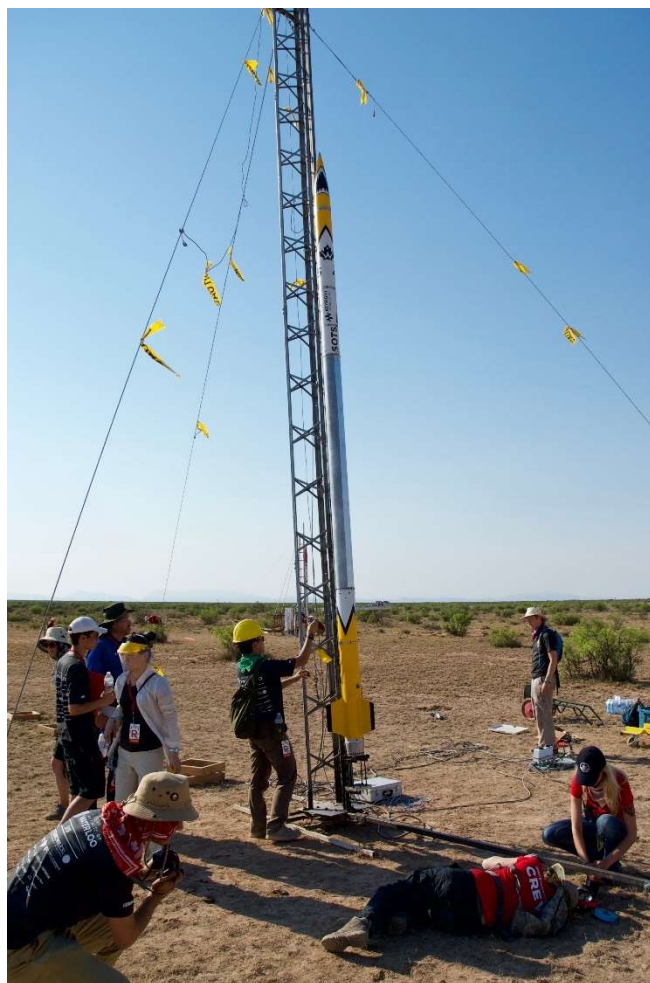


Figure 1: Shark of the Sky on the launch tower.

1.1 Concept of Operations

When the team reaches the launch site at competition a full day is spent setting up launch infrastructure. Starting early the next morning at 2:00 am a crew arrives at the launch site and begins final assembly of the rocket. This is completed at approximately 7:00 am, at which point the rocket is transported in two

pieces, first to the judges to receive approval for flight, and then out to the launch tower. The two pieces are then bolted together, and the completed rocket is placed onto the launch tower. The rocket and launch tower are then raised together.

At this point the rocket may need to wait on the launch tower, potentially for many hours, until the team is given its turn to launch. Pre-launch operations are then completed, filling the rocket with propellant, and arming the recovery pyrotechnics. As shown in Figure 2, the engine is then fired, propelling the rocket into the air, and accelerating the rocket to near the speed of sound before burning out 15 seconds later. From there the rocket coasts up until reaching its highest point, apogee. The Parachute is then deployed in a reefed (partially open) state. When the rocket descends through 1,500 ft the parachute is then fully deployed, slowing the rocket for a gentle touchdown where it is then tracked by a recovery team and recovered.

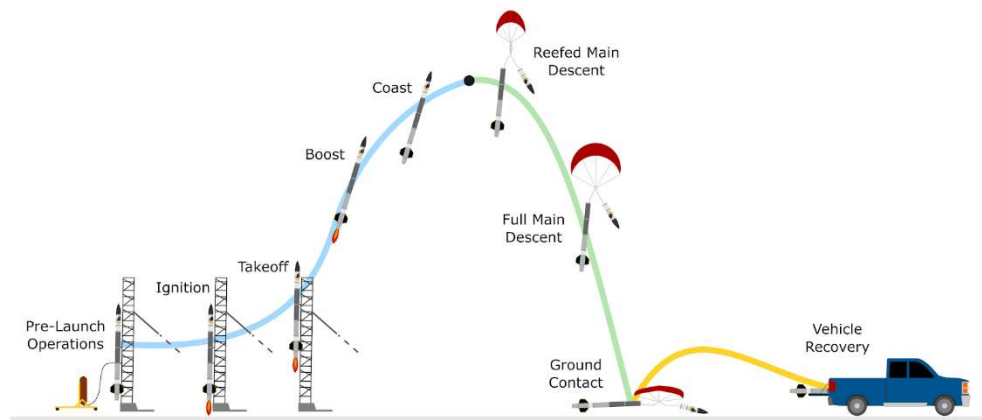


Figure 2: Kraken of the Sky Concept of Operations.

1.2 Recovery System Functionality

The recovery system is based around the concept of a reefed parachute. When the rocket reaches apogee two small gunpowder charges are set off, puncturing CO₂ canisters, and pressurizing the recovery bay. This causes the pins which retain the nose of the rocket to shear, blowing open the recovery section and ejecting the parachute.

The parachute is now deployed but is prevented from fully opening by a cord running its circumference known as a reefing line. The partially inflated parachute slows the fall rate of the rocket to ~35 m/s, allowing it to fall from 30,000 ft to 1,500 ft quickly and hence preventing it from being blown far off course. Once the rocket hits 1,500 ft another pyrotechnic charge is triggered, severing the reefing line, and slowing the decent rate of the rocket to 9 m/s so that it is not damaged when it then hits the ground.

The pyrotechnics used to deploy and dis-reef the parachute are triggered using a pair of redundant and dissimilar commercial off the shelf (COTS) altimeters. The use of COTS altimeters is required by the competition to reduce the risk of pyrotechnics being misfired and causing injury, as well as ensuring teams cannot tamper with their maximum recorded altitude. The first is a Perfectflight Stratologger, which is a relatively simple altimeter, but is robust and provides live altitude data in flight via UART [2]. The second altimeter is a Featherweight Raven 4, which is a high end altimeter that logs additional data such as

horizontal and vertical acceleration [3]. The altimeters are powered with two independent 9V batteries and are switched on using magnetic switches (mag switches) by waving a magnet over the outside of the rocket.

Previously, Waterloo Rocketry's rockets have used a more common two parachute system, where a small parachute, known as the drogue parachute, is deployed at apogee, followed by a larger main parachute being deployed at 1,500 ft. As such most documentation refers to the two sets of pyrotechnics as the drogue pyros and the main pyros. A block diagram of the 2019 electronics are shown in Figure 3.

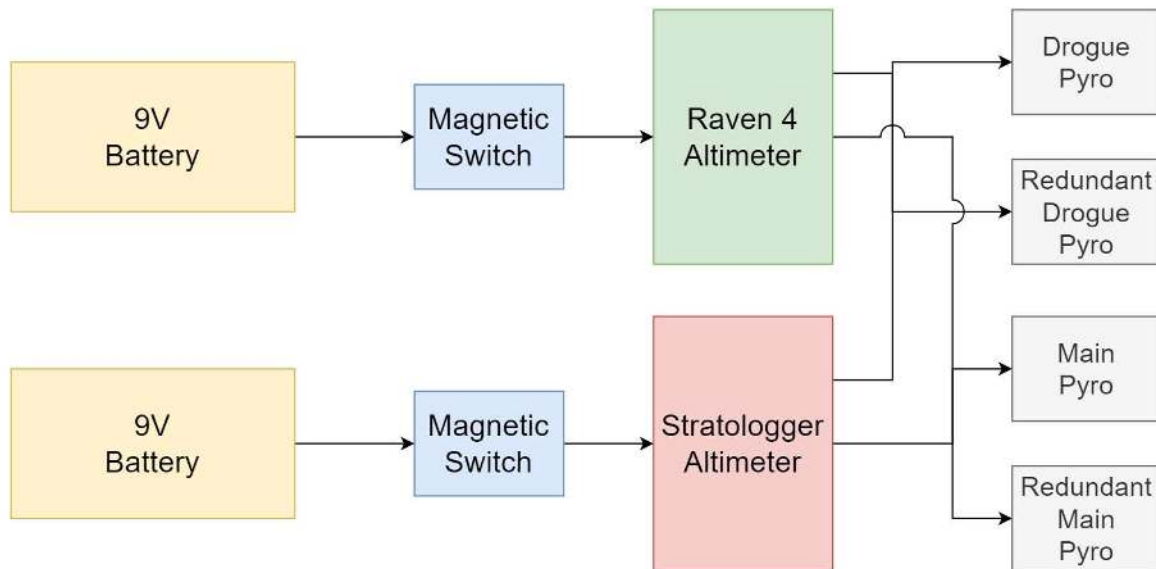


Figure 3: 2019 Recovery electronics block diagram.

1.3 Current Issues

While the current recovery electrical system is relatively robust. It is quite basic and leaves something to be desired in terms of safety and feedback. The recovery electronics are only armed once the rocket is vertical on the launch tower. To do this the operator must climb a step ladder and use a long pole with a magnet taped to it to turn on the mag switches, essentially batting it at the rocket until they hear the buzzers of the altimeters. While not a critical safety risk, this is not ideal, as there is a risk of the pyrotechnics being fired whenever the altimeters are armed, and if they were to fire, there is a chance the operator could be injured. In addition, there is a risk of falling off the step ladder as it is a somewhat precarious procedure.

Another downside to the current system is that there is no form of feedback about the state of the system aside from the buzzers built into the altimeters. This means that there is no way to confirm immediately prior to launch that the system is still fully armed and increases the risk of issues such as incorrect wiring or accidentally installing dead batteries.

1.4 Objective

To mitigate issues with the current recovery electrical system a project was proposed to create a remote arming system which would use the rocket's pre-existing CAN bus [4] and ground communication network to interface with the recovery electronics. The objective of the remote arming system is to create a system that allows the recovery pyrotechnics of Waterloo Rocketry's rockets to be safely armed by remotely powering them up from a safe distance.

2.0 Problem Definition

The defining problem for the Remote Arming system, as shown in Figure 4, is to create a system which always fails safe. Most systems on the rocket need to fail safe, however usually this has a well defined, single meaning. For example, the vent valve on the rocket's oxidizer tank should always fail open, since this ensures all the oxidizer will vent out without issue. There is no situation where the vent valve failing open will cause added risk. However, for the Remote Arming System, failing safely does not always mean the same thing. While the rocket is on the ground, the remote arming system needs to fail in the disarmed state, as if it fails armed, the pyrotechnics could accidentally be triggered, and an operator could be injured. However, once the rocket launches, the altimeters must stay armed, or the rocket's parachute may not deploy, and it would slam into the ground at hundreds of kilometers per hour, obliterating the rocket, and destroying anything it might land on. As such once in flight failing safely means ensuring the system fails armed.

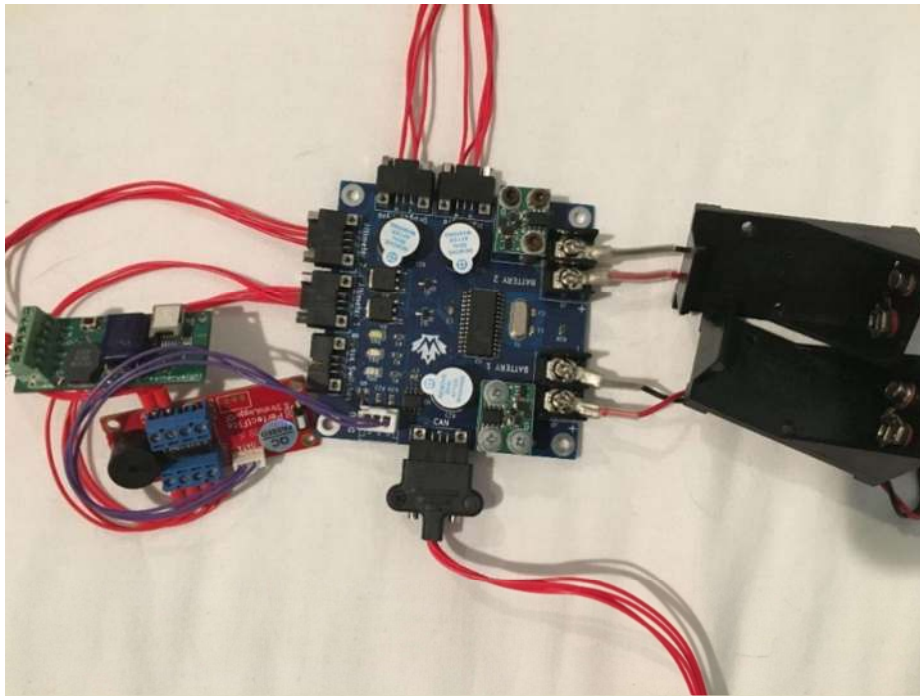


Figure 4: Complete Remote Arming system.

2.1 Rules and Regulations

Waterloo Rocketry is currently involved with two different competitions: The Spaceport America Cup, and Launch Canada. Each of these competitions require that teams comply with their Design Test and Evaluation Guides (DTEGs) [5] which layout the rules for the design and testing of all aspects of team's rockets. Since Launch Canada is based off the Spaceport America Cup, the Launch Canada DTEG is a more updated and comprehensive set of rules, such that for the most part, if the team complies with the Launch Canada DTEG, the rocket will also be compliant with the Spaceport America Cup DETG. The Launch Canada regulations applicable to the remote arming system are summarized in Table 1 [5].

Table 1: Applicable Launch Canada DTEG requirements.

Requirement Number	Requirement
R4.8.1	Launch vehicles shall implement completely independent and redundant recovery systems, including: • Arming switch • sensors/flight computers • power supply • energetics • "electric initiators"
R4.8.4	At least one redundant recovery system electronics subsystem shall implement a proven COTS flight computer (e.g., StratoLogger, G-Wiz, Raven, Parrot, AIM, EasyMini, TeleMetrum, RRC3, etc.).
R4.9.1	To be acceptable for use as part of a recovery system at the competition, SRAD flight computers shall be thoroughly ground tested.
R4.9.2	SRAD flight computers should additionally be flight tested prior to the competition, for example as a payload on a smaller vehicle
R5.2.1	All energetics shall be safed until the rocket is in the launch position, at which point they may be "armed"
R5.2.2	All energetics that could create a hazard to personnel if inadvertently actuated shall only be armed remotely when all personnel have been cleared to a safe distance
R5.2.4	All energetic device arming features shall be located on the airframe such that any inadvertent energy release by these devices will not impact personnel arming them.
R6.1	Electrical assemblies and devices shall be compatible with the external and self-induced electromagnetic environments that will exist during flight or testing.
R6.3	Onboard avionics and control systems shall be electrically isolated from any pyrotechnic or electro-explosive devices such that a short of a pyrotechnic device cannot disable the control system.
R6.4	Electrical systems shall be designed to limit or prevent a short in one system from disabling other flight- or safety-critical systems.

2.2 Constraints and Criteria

In addition to the Rules and Regulations imposed by the competitions, there are several constraints and criteria that the system must meet to be an effective part of Waterloo Rocketry's rocket. These are summarized in Table 2.

Table 2: Remote Arming criteria and constraints.

Criteria
The system shall be able to handle the current required to fire the two redundant pyrotechnics at the same time (10A for at least 1 second).
The system shall fit within the 6" body of the rocket.
The system shall be easy to bypass should an issue be found with it, allowing the recovery electronics to be operated in the same set up used in previous rockets.
Constraints
The system should be made as light and compact as possible to ensure minimum impact on vehicle performance.
The system should provide telemetry data on the state of the recovery system, allowing its state to be easily monitored.
The system should use positively locking electrical connection methods to ensure the risk of electronics being detached under high Gs and vibration is low.
It should be possible to tell the state of the system via auditory cues to reduce the risk of personnel inadvertently approaching the armed system.
The system should be optimized to survive being powered on in the rocket in the hot desert for many hours in the event of delays due to other teams attempting to launch.

3.0 System Design

3.1 Overview

The remote arming system is a single printed circuit board assembly that is installed between the recovery batteries and the altimeters. Roughly speaking the Remote Arming System can be split into three main sections: The power management circuitry, the altimeter control circuitry, and the microcontroller and telemetry circuitry as seen in Figure 5. It is all controlled using a PIC18F26K83 microcontroller and is designed with similar hardware to the rest of Waterloo Rocketry's RocketCAN system, including using the team's custom CAN library for handing communications. [4]

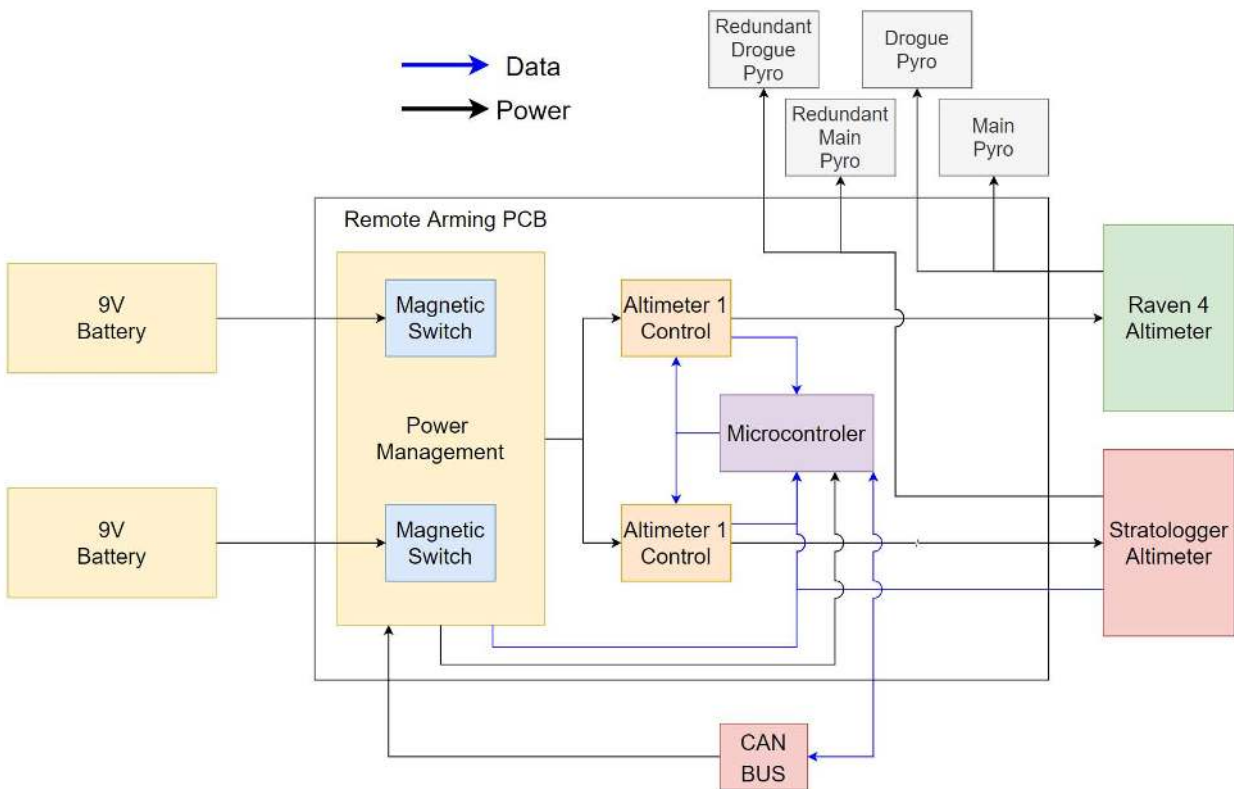


Figure 5: Remote Arming block diagram.

3.2 Failing Safely

As previously mentioned, a key challenge for the remote arming system is designing a system which fails disarmed on the ground, but always fails armed in flight. Unfortunately, it is not possible to design a system which will always fail armed yet also always fail disarmed, so a compromise was made. The system has been designed such that it will always fail armed. However, a pair of mag switches ensure that the recovery system can always be manually disarmed in the same way that the altimeters have been armed in previous years. In addition, buzzers which are continuously on whenever the altimeters are powered ensure that there is no possibility personnel approaching the system without knowing it is armed.

Since both arming circuits are controlled by a single microcontroller, several measures have been taken to ensure that it does not cause a single point of failure:

- As shown in Figure 9, there is a pull up resistor which ensures that if the microcontroller fails or loses power, the circuit will keep the altimeter powered.
- The code on the microcontroller ensures that its default state on power up is armed, as such any temporary loss of power will not result in the altimeters disarming.
- A watchdog timer is used, such that if main loop is ever hung up, the microcontroller will be reset.
- The microcontroller is powered both by the recovery batteries, and the Rocket's CAN bus, which adds redundancy and reduces the chances of a brown out.

3.3 Concept of Operations

Since the logic of Remote Arming is designed to default to armed, some care must be taken in setting up the system for launch. The rough procedure is as follows:

- Since the mag switch's default state is on, during assembly the 9V batteries should be installed first, the mag switches turned off, and then the pyrotechnics should be attached. This is true regardless of remote arming.
- Once the rocket has been full assembled, inspected, and loaded onto the launch rail, but before the rail is raised, the CAN bus should be powered up, Remote Arming should be commanded to disarm the altimeters, and then a magnet should be used to turn on the mag switches, checking the remote arming telemetry to ensure both mag switches are functioning correctly. The indicator buzzer will also play a repeating tone to indicate both batteries are functioning correctly.
- The CAN bus can then be powered down and Remote Arming will continue operating using recovery battery power.
- Once the rocket is vertical on the tower and ready for launch, Remote Arming will be commanded to arm. The telemetry will then be checked to ensure the system is ready for launch. This includes the following:
 - Both battery and mag switch voltages are good.
 - Remote Arming reports both altimeters armed.
 - Current draw from the recovery batteries matches expected value for all systems armed.
 - Drogue and Main pyro voltages for each altimeter read as expected for armed state.
 - Altitude data is being received from the Stratolger altimeter.

3.4 Power Management

Remote arming is powered by two 9V batteries, each with their own voltage sensing and mag switch, as shown in Figure 6. A SM74611 smart diode is used with each battery to ensure a difference in voltage between the two does not cause issues. These diodes are designed for solar power generation and have a nominal voltage drop of only a few mV thanks to an internal FET and charge pump circuit, ensuring there is minimal power loss through the diode.

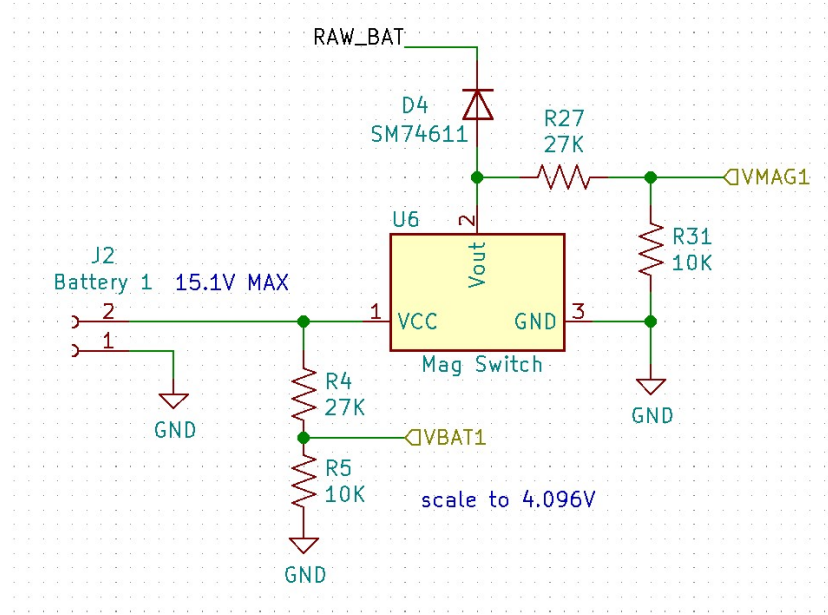


Figure 6: 9V battery management circuit.

The microcontroller is powered by the CAN bus via the circuit shown in Figure 7. A polyfuse is used to ensure that if there is a short in the Remote Arming system it will not bring down the rocket's entire control system. An INA180 amplifier is used with a shunt resistor to allow Remote Arming to measure how much current is being drawn from the CAN bus. Finally, a CUS08F30 Schottky diode is used to prevent Remote Arming from powering the CAN bus with its backup 5V regulator which draws power from the 9V batteries when the CAN bus is turned off to save power.

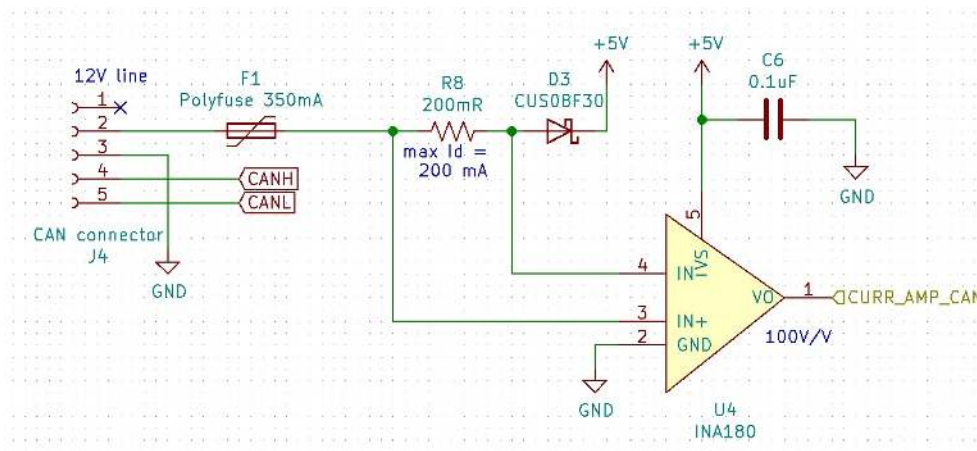


Figure 7: CAN bus connector and current sensing circuit.

Current sensing from the 9V batteries is achieved using another INA180 amplifier as shown in Figure 8. Two shunt resistors are used in parallel to ensure single fault tolerance. If one resistor is damaged, the current reading will be inaccurate, but the altimeters will still be powered.

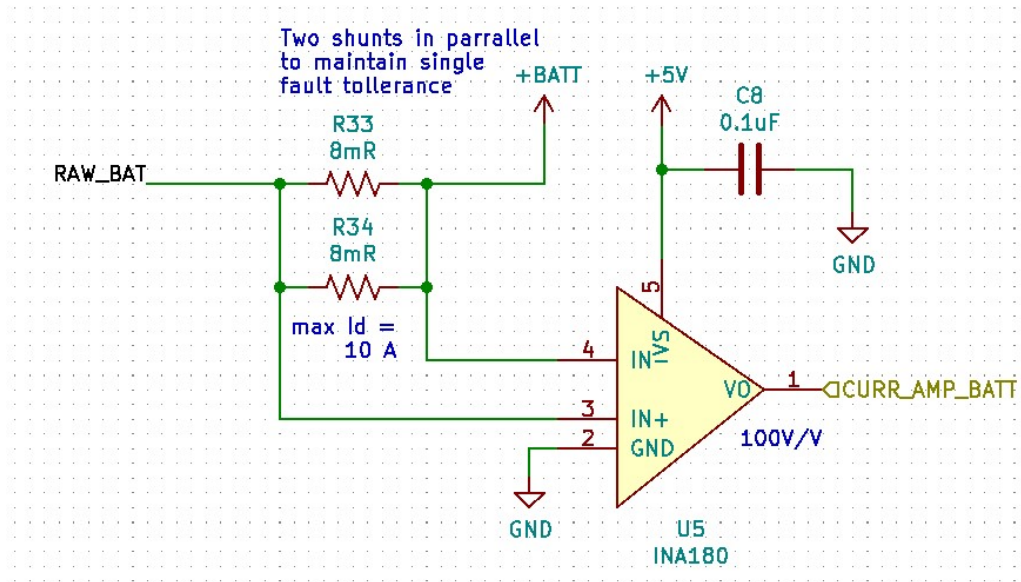


Figure 8: Recovery battery current sensing circuit.

3.5 Altimeter Control

As seen in Figure 9, power to each altimeter is controlled by an AOD417 P-channel MOSFET, which is in turn controlled by a DMG2302UK N-channel MOSFET to allow the circuit to be controlled with the microcontroller's 5V logic. This high side switching is slightly more complex than if low side switching was implemented with a single N-channel MOSFET but ensuring that the altimeters and Remote Arming has a shared ground is needed for reading altitude data from the Stratologger. This MOSFET combination has already been flow as part of the injector valve control board in 2019 without issue and continued to function even after the rocket landed under drogue and experienced significant shock loading, so there are no concerns about its robustness. The pull up solder jumpers are for use with the Raven 4 altimeter which does not have a dedicated positive pin for the drogue or main pyrotechnics, and as such requires that they be externally connected to power. As mentioned previously, each power control circuit has a buzzer which is on whenever the altimeter is powered. Two different frequencies are used to make it possible to identify which altimeters are powered.

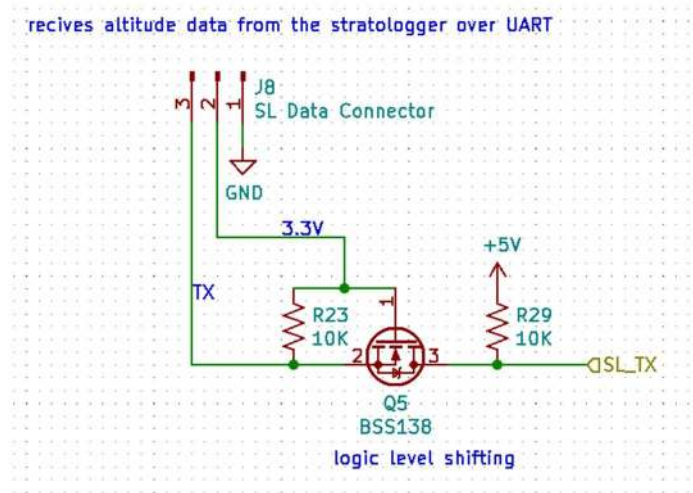


Figure 10: Stratologger telemetry level shifting circuit.

3.7 Testing to Date

As of present the Remote Arming system has undergone full system testing under benchtop conditions. It has also undergone a series of simulated flights where the complete recovery electronics system was put in a vacuum chamber. A partial vacuum was pulled to simulate an increasing altitude, then air was slowly introduced back in to simulate apogee and decent. Referencing an aircraft altimeter, it was confirmed that the drogue channel fired at “apogee”, and the main channel fired at a reading of 1500 ft, as well as that Remote Arming reported all the correct telemetry. Unfortunately, this test only works with the Stratologger altimeter since its altitude measurement is solely barometer based, while the Raven 4 uses an accelerometer along with the barometer. Both the Raven and Stratologger altimeters were tested with Remote Arming by connecting the to a computer and using their respective simulated flight tests.

4.0 Conclusions

4.1 Conclusions

Waterloo Rocketry is currently finishing the design and fabrication of its next rocket, Kraken of the Sky. As part of this rocket a new Remote Arming system has been developed. The objective of this system is to remove the need for personnel to climb a ladder and use a pole to arm the rocket's recovery pyrotechnics and therefore improve the safety of Waterloo Rocketry's launch operations. A secondary objective of the system was to provide telemetry data on the state of the recovery system, allowing the system's health to be monitored up to and during launch. To date the system has been designed, manufactured, and tested, including manufacturing a second revision which fixes a critical bug, and added battery voltage and current sensing.

4.2 Next Steps

While Remote Arming has undergone extensive benchtop development and testing, as well as simulated flights in a vacuum chamber, to date the system has not yet been tested with real pyrotechnics, or with the complete set of boards which will be used on the rocket's CAN bus. Upcoming testing includes the following:

- A verification test will be done to ensure that the system is resistant to a short-circuited pyrotechnic.
- There will be a series of complete CAN bus system integration tests to ensure the system works properly as a whole.
- The Remote Arming system will be used during the recovery drop test, where the recovery system is dropped from a helicopter, testing the system under flight-like conditions.

In addition, while the Remote Arming software development is largely out of the scope of this system, there is still notably work to do in developing the error reporting, and state estimation algorithms which will run on the Remote Arming system.

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